New Zealand Land Use and Management classification system v0.4.1

Status Draft **Website** https://nzlum.landcareresearch.co.nz/ **Git repository** https://github.com/manaakiwhenua/nzsluc

Scheme IRI https://nzlum.landcareresearch.co.nz/landuse/scheme/nzlum0.4.1

> ***"Kotahi te aho ka whati, ki te kāpuia e kore e whati."***
> ***"One strand of flax is easy to break, but many strands together will stand strong."***
>
> — Kīngi Tāwhiao

Primary code	Primary class	Primary colour	Secondary code	Secondary class	Secondary colour	Tertiary code	Tertiary class	Tertiary colour
1.0.0	**Conservation and Minimal Use**	#7B6DB5	1.1.0	**Nature conservation**	#6A51A3	1.1.1	Strict nature reserve	#3F007D
						1.1.2	Wilderness area	#54278F
						1.1.3	National Park	#6A51A3
						1.1.4	Natural feature protection	#807DBA
						1.1.5	Habitat or species management area	#9E9AC8
						1.1.6	Protected landscape	#BCBDDC
						1.1.7	Other conserved area	#DADAEB
			1.2.0	**Cultural and natural heritage**	#AD2D4A	1.2.1	Indigenous cultural heritage	#870330
						1.2.2	Cultural heritage	#AD2D4A
						1.2.3	Natural heritage	#DD5A70
			1.3.0	**Minimal use from relatively natural environments**	#FFC77A	1.3.1	Surface water supply	#50816D
						1.3.2	Ground water	#79AB96
						1.3.3	Grazing native vegetation	#FFD5B1
						1.3.4	Production from indigenous vegetation	#C6894C
						1.3.5	Customary use	#A36B2E
						1.3.6	Defence land	#FF9E17
						1.3.7	Environmental & infrastructure protection	#FBB979
						1.3.8	Carbon forest	#2E5E4B
			1.4.0	**Unused and transitioning land**	#C8C69F	1.4.1	Rehabilitating land	#B2AC8B
						1.4.2	Unused land	#DFD8B6
2.0.0	**Production agriculture and plantations**	#6E8C44	2.1.0	**Plantation forests**	#122112	2.1.1	Exotic plantation forestry	#0F3D1A
						2.1.2	Indigenous plantation forestry	#23502B
						2.1.3	Other production uses	#37633D

Primary code	Primary class	Primary colour	Secondary code	Secondary class	Secondary colour	Tertiary code	Tertiary class	Tertiary colour
						2.1.4	Planted environmental & infrastructure protection	#4A774F
			2.2.0	**Grazing modified pasture systems**	#769B5C	2.2.1	Dairy	#5D8144
						2.2.2	Intensive dry stock	#769B5C
						2.2.3	Extensive dry stock	#91B675
			2.3.0	**Short-rotation and seasonal cropping**	#C6A14D	2.3.1	Arable cropping	#A98734
						2.3.2	Arable and mixed livestock cropping	#C6A14D
						2.3.3	Short-rotation horticulture	#E3BC67
						2.3.4	Seasonal flowers and bulbs, and turf farming	#FED57E
			2.4.0	**Perennial horticulture**	#9EA083	2.4.1	Tree crops	#818367
						2.4.2	Vine fruits	#9EA083
						2.4.3	Other perennial horticulture	#BBBEA0
			2.5.0	**Intensive horticulture**	#4A4C39	2.5.1	Production nurseries	#565126
						2.5.2	Glasshouses/shadehouses	#797346
			2.6.0	**Intensive animal production**	#C79588	2.6.1	Intensive animal containment	#9A6C60
						2.6.2	Poultry farms	#C79588
						2.6.3	Piggeries	#B08074
						2.6.4	Horse studs	#DDAA9D
						2.6.5	Aquaculture	#F5C0B2
			2.7.0	**Water and wastewater**	#528582	2.7.1	Stock water	#528582
						2.7.2	Effluent pond	#205351
						2.7.3	Water treatment – land application	#396B69
						2.7.4	Water treatment – wetland	#6C9F9C
						2.7.5	Irrigation reservoirs and canals	#86BAB7
			2.8.0	**Vacant and transitioning land**	#B6C8BC	2.8.1	Rehabilitating land	#B6C8BC
						2.8.2	Abandoned land	#CADCD0
						2.8.3	Degraded land	#94A599
						2.8.4	No defined use	#DBEDE0
			2.9.0	**Smallholder farm**	#E5D64F			
3.0.0	**Built environment**	#8E7A78	3.1.0	**Residential**	#815E5B	3.1.1	High-density residential	#4C3B39
						3.1.2	Medium-density residential	#644E4C
						3.1.3	Low-density residential	#815E5B
						3.1.4	Rural residential	#A87A76
			3.2.0	**Public recreation and services**	#C19878	3.2.1	Outdoor recreation	#C19878
						3.2.2	Indoor recreation	#958273

Primary code	Primary class	Primary colour	Secondary code	Secondary class	Secondary colour	Tertiary code	Tertiary class	Tertiary colour
						3.2.3	Community services	#E09962
			3.3.0	**Commercial**	#6E94BA	3.3.1	Retail	#6E94BA
						3.3.2	Office	#396285
						3.3.3	Hospitality	#A4CAF2
						3.3.4	Entertainment	#89AFD6
						3.3.5	Healthcare	#D0E5FC
						3.3.6	Transportation & warehousing	#547B9F
			3.4.0	**Manufacturing and industrial**	#9598A8	3.4.1	General purpose factory	#9598A8
						3.4.2	Food processing factory	#CED1E2
						3.4.3	Major industrial complex	#838695
						3.4.4	Sawmill	#A8AABB
						3.4.5	Farm buildings/infrastructure	#E2E4F6
						3.4.6	Abattoirs	#BBBDCE
			3.5.0	**Utilities**	#955085	3.5.1	Fuel powered electricity generation	#7652A3
						3.5.2	Hydroelectricity generation	#9670C4
						3.5.3	Wind electricity generation	#D6B2FD
						3.5.4	Solar electricity generation	#B78FE5
						3.5.5	Electricity substations and transmission	#955085
						3.5.6	Gas treatment, storage, and transmission	#B36CA2
						3.5.7	Water extraction and transmission	#D289BF
						3.5.8	Navigation and communication	#F2A6DE
			3.6.0	**Transportation**	#72737E	3.6.1	Airports/aerodromes	#75777E
						3.6.2	Roads	#494B51
						3.6.3	Railways	#33353A
						3.6.4	Ports and water transport	#61636A
			3.7.0	**Mining**	#E1CEA7	3.7.1	Mines	#836639
						3.7.2	Quarries	#9E7E51
						3.7.3	Tailings	#BA9869
						3.7.4	Evaporation basins	#FCD7A5
						3.7.5	Extractive industry not in use	#D7B383
			3.8.0	**Waste treatment and disposal**	#E77148	3.8.1	Landfills	#AC3F25
						3.8.2	Transfer stations and recycling facilities	#E97453
						3.8.3	Municipal wastewater	#CB593B

Primary code	Primary class	Primary colour	Secondary code	Secondary class	Secondary colour	Tertiary code	Tertiary class	Tertiary colour
						3.8.4	Wastewater treatment – land application	#FF9475
						3.8.5	Stormwater management	#FEBDAA
			3.9.0	**Vacant and transitioning land**	#D69B97	3.9.1	Brownfield development	#87636E
						3.9.2	Greenfield development	#AC8792
						3.9.3	Idle and derelict land	#D3ACB7

Class definitions

1.0.0 Conservation and Minimal Use #7B6DB5

Land that has a relatively low level of human intervention. The land may be formally reserved by government for conservation purposes, or conserved through other legal or administrative arrangements. Areas may have multiple uses but nature conservation is a central consideration. (Some land may be unused because of a deliberate decision of the government or landowner, or due to circumstances).

Scope note Areas with lower levels of biodiversity protection are anticipated to have other primary uses, but biodiversity protection ranking could be captured as an ancillary use with class 1.1.4 and 1.1.5.

Usage note Where a classification is based on information about legal protection, the relevant information (e.g. the Act) should be mentioned in the `comment` field.

1.1.0 Nature conservation #6A51A3

Nature conservation classes are determined by the classification scheme adopted for the Protected Areas Network of New Zealand (PAN-NZ).

1.1.1 Strict nature reserve #3F007D

Protected area managed mainly for science. An area of land possessing outstanding or representative ecosystems, geological or physiological features or species, which is available primarily for scientific research and/or environmental monitoring.

Source: IUCN (2008) Protected Areas Management Categories, Category Ia: strict nature reserve.

1.1.2 Wilderness area #54278F

Protected area managed mainly for wilderness protection. A large area of unmodified or slightly modified land, retaining its natural character and influence, without permanent or significant habitation, which is protected and managed so as to preserve its natural condition.

Source: IUCN (2008) Protected Areas Management Categories, Category Ib: wilderness area.

1.1.3 National Park #6A51A3

Protected area managed mainly for ecosystem conservation and recreation. A natural area of land, designated to: a) protect the ecological integrity of one or more ecosystems for the current and future generations; b) exclude exploitation or occupation detrimental to the purposes of designation of the area; and c) provide a foundation for spiritual, scientific, educational, recreational and visitor opportunities, all of which must be environmentally and culturally compatible.

Source: IUCN (2008) Protected Areas Management Categories, Category Ib: national park.

1.1.4 Natural feature protection #807DBA

Protected area managed for conservation of specific natural features. An area containing one or more specific natural or cultural features that are of outstanding value because of their inherent rarity, representative or aesthetic qualities, or cultural significance.

Source: IUCN (2008) Protected Areas Management Categories, Category III: Natural Monument or Feature.

1.1.5 Habitat or species management area #9E9AC8

Protected area managed mainly for conservation through management intervention. An area of land or sea subject to active intervention for management purposes to ensure the maintenance of habitats or to meet the requirements of specific species. This may include areas on private land.

Source: IUCN (2008) Protected Areas Management Categories, Category IV: Habitat/Species Management Area.

1.1.6 Protected landscape #BCBDDC

Protected area managed mainly for landscape conservation and recreation. An area of land where the interaction of people and nature over time has produced a distinct character with significant aesthetic, cultural or ecological value, and often high biodiversity.

Source: IUCN (2008) Protected Areas Management Categories, Category V: Protected Landscape/ Seascape.

1.1.7 Other conserved area #DADAEB

Land under forms of nature conservation protection that fall outside the scope of the CAPAD classification. This includes heritage agreements, voluntary conservation arrangements, registered property agreements, and recreation areas with primarily native cover .

Scope note If the recreation area is primarily cleared or non-native cover assign to 3.2.1 ‘Outdoor recreation’

1.2.0 Cultural and natural heritage #AD2D4A

Conservation classes are for purposes other than biodiversity protecton.

1.2.1 Indigenous cultural heritage #870330

Indigenous cultural heritage sites and areas (e.g. historical pā sites, New Zealand land wars sites, and other indigenous heritage sites).

1.2.2 Cultural heritage #AD2D4A

Cultural heritage areas (e.g. historical mining sites, selected DOC 'iconic sites', Toru Whenua and other cultural landmarks).

1.2.3 Natural heritage #DD5A70

Protected areas for the conservation of specific natural features and landscapes (e.g. UNESCO Global Geoparks).

1.3.0 Minimal use from relatively natural environments #FFC77A

This class includes land that is subject to relatively low levels of intervention or that is largely unused in the context of prime use or use for resource protection. This land may be covered with indigenous or exotic plant species. It includes land where the structure of the native vegetation generally remains intact despite deliberate use, although the floristics of the vegetation may have changed markedly (e.g. grazing on native tussock land).

Scope note Where native grasses have been deliberately and extensively replaced with other species, the land use should not be classified under class 1.

1.3.1 Surface water supply #50816D

An area managed as a catchment for water supply.

1.3.2 Ground water #79AB96

An area managed as an aquifer recharge zone.

1.3.3 Grazing native vegetation #FFD5B1

Land uses based on grazing by domestic stock on native vegetation where there has been limited or no deliberate attempt at pasture modification (e.g. native tussock grazing). Some change in species composition may have occurred.

Usage note This is probably limited to South Island high-country farms. Verification (e.g. assessment of vegetation on grazed land) to assess the extent of modification is required. ALUM specifies this class when there is greater than 50% dominant native species, although this criterion needs to be confirmed as being appropriate in a New Zealand context.

1.3.4 Production from indigenous vegetation #C6894C

Commercial production from retained (non-plantation) native forests and related activities on public and private land (e.g. selective native timber production without clearfelling, and other non-sawlong, non-pulpwood, native forest products including oil, wildflowers, firewood, fenceposts, and mānuka/kānuka honey).

Change note The definition of this class was clarified in NZLUM v0.4.0 by explicitly referring to non-plantation native forests.

1.3.5 Customary use #A36B2E

Natural environments associated with traditional and sustainable Indigenous food-gathering practices (mahinga kai).

Scope note This may include land zoned for Māori purposes, and should only describe contemporary, not historical, use.

Usage note This is often an ancillary use; assign only if customary use is the prime use of the land.

1.3.6 Defence land #FF9E17

Natural areas allocated to field training, weapons testing, and other field defence uses, predominantly in rural areas.

Scope note Areas associated with buildings or more built environments on defence land are captured under an urban class.

Example Kaipara Air Weapons Range

Example Waiouru Military Camp

1.3.7 Environmental & infrastructure protection #FBB979

Land, usually under vegetative cover, used for non-production or environmental purposes (e.g. prevention of land degradation, windbreaks, shade, shelter). Includes land with a primary purpose of flood management (e.g. stop banks, spillways).

Scope note This is not limited to indigenous vegetation.

Scope note Land planted for environmental and infrastructure protection should be captured under 2.1.4.

1.3.8 Carbon forest #2E5E4B

Retained (non-planted) indigenous vegetation set aside for carbon credits.

1.4.0 Unused and transitioning land #C8C69F

Land unused for rural production and urban development, without particular conservation instruments, and without any apparent built structure; or land that is transitioning towards a vacant and considerably natural state.

Scope note Corridors and roadside areas may fit under this class.

Scope note Unused land (in the sense of non-productive, non-conservation or non-built uses), such as cliffs, rock faces, boulders, and tors, and where there are relatively low levels of disturbance, may be included.

Scope note Does not per se include land undergoing natural succession in the context of changing plant species.

Change note https://github.com/manaakiwhenua/nzsluc/issues/36

1.4.1 Rehabilitating land #B2AC8B

Degraded land (e.g. due to erosion or flood damage) that is being actively rehabilitated through planting with indigenous or exotic species to return land to a natural state.

Scope note This may include riparian plantings.

Change note https://github.com/manaakiwhenua/nzsluc/issues/36

1.4.2 Unused land #DFD8B6

Land that is unused or unusable for productive agriculture or urban uses, such as cliffs, rock faces, boulders, and tors, where there are relatively low levels of disturbance.

Example A vacant farm, apparently undergoing natural succession.

Example CERA Red Zone land in Christchurch, abandoned after the 2011 Canterbury earthquake and presently pending use.

Change note https://github.com/manaakiwhenua/nzsluc/issues/36

2.0.0 Production agriculture and plantations #6E8C44

Land used principally for primary production, where native vegetation has largely been replaced by introduced species through clearing, the sowing of new species, the application of fertilisers, or the dominance of volunteer species. Includes plantation forests, pasture production for stock, cropping and fodder production, and horticultural production.

Scope note If there is evidence of irrigation infrastructure or water-take consents, record irrigation as a management practice, even if irrigation water has not been recently applied.

Usage note Fallow or ploughed land should be assigned to the most likely land use based on the dominant activity conducted in comparable nearby areas or other available evidence. Fallow or ploughed land should be allocated to the relevant pasture, cropping or horticultural class (rather than using land in transition). The fallow or ploughed status should be recorded in the management field.

2.1.0 Plantation forests #122112

Land on which plantations of trees or shrubs (native or exotic species) have been established (i.e. planted) for production, or environmental and resource protection purposes, including carbon forestry.

Scope note This includes farm forestry and may consist of monocultures or mixed species.

2.1.1 Exotic plantation forestry #0F3D1A

An area managed for pulpwood or saw-log production, or for carbon sequestration and carbon credits (exotic species).

Usage note The specific commodity or commodities should be noted if known.

Usage note Specific additional attributes that could be captured in comments or supplementary attributes are plantation age, rotation number, and species.

Change note Consideration of carbon sequestration and carbon credits were added in NZLUM v0.4.0

2.1.2 Indigenous plantation forestry #23502B

An area managed for pulpwood or saw-log production, or for carbon sequestration and carbon credits (native species).

Usage note The specific commodity or commodities should be noted if known.

Usage note Specific additional attributes that could be captured in comments or supplementary attributes are plantation age, rotation number, and species.

Change note Consideration of carbon sequestration and carbon credits were added in NZLUM v0.4.0

2.1.3 Other production uses #37633D

An area managed for non-pulpwood production, including oil, wildflowers, and honey plantations.

Usage note The specific commodity or commodities should be noted if known.

2.1.4 Planted environmental & infrastructure protection #4A774F

An area managed for environmental and indirect production uses (e.g. prevention of land degradation, windbreaks, shade, and shelter) where trees or shrubs have been planted.

2.2.0 Grazing modified pasture systems #769B5C

Grazing pasture and/or forage systems based on significant active modification or replacement of the natural vegetation.

Scope note Land under pasture at the time of mapping may be in a rotation system, so that at another time the same area may be, for example, under cropping.

Scope note Dairy support is land that is used to support non-lactating dairy stock (dry cows, heifers and calves). It will include any feed required, and will often include winter crops and potentially summer crops (location/irrigation dependent), along with cereal crops, such as maize, barley, and wheat. It can also include feed that is cut and carried to the milking platform. Dairy support land that is not actively used for grazing should be classified as an arable use.

Usage note The ability to distinguish between dairy and dry-stock production is provided by including the relevant commodity information, such as `cattle dairy`, `cattle beef`, `sheep wool`, `sheep meat`, etc. Multiple commodities should be recorded if appropriate. Crops used in rotation should also be recorded as commodities, if known.

2.2.1 Dairy #5D8144

Land on which milking cows (or other stock, such as goats or sheep) are grazed during the milking season, where dairy is the primary purpose of the land.

Scope note Dairy production systems can include rotations of grazed forage crops and maize for silage, and dry-stock grazing, but this class should only be used where dairy is the primary purpose of the land.

Scope note Where the land is permanently used for dry-stock grazing, it should be classified under dry-stock land use.

2.2.2 Intensive dry stock #769B5C

Non-milking platform grazing land with higher inputs (e.g. fertiliser), higher likelihood of irrigation, and higher stocking rates; typically on flat or rolling terrain.

Scope note Land used for high-intensity dry-stock grazing may include rotations for arable or winter forage crops, as well as grazing of non-lactating (dry) dairy cattle, beef cattle, sheep, and cattle breeding.

Usage note Stock density is an ideal attribute to assist in classification of intensive vs extensive dry stock; physiography (i.e. flat/hill/high country) can be used as a proxy where stock density data is absent.

Change note A prior definition of this classes treated deer, goats, and horses as extensive by definition; it is now recogised that such systems can also be intensive.

Change note https://github.com/manaakiwhenua/nzsluc/issues/66

2.2.3 Extensive dry stock #91B675

Grazing on modified pastures with relatively fewer inputs, lower likelihood of irrigation, and lower stocking rates; more likely on hill, hard-hill, or high-country terrain.

Scope note Grazing livestock other than dairy, sheep or beef should usually be captured in this class (though it does not exclude sheep or beef)

Scope note Where there is a high proportion of indigenous vegetation for grazing land, land use should be classified as grazing native vegetation.

Scope note Arable or winter forage crops are unlikely to be common rotations in this land-use category

Usage note The specific commodity or commodities should be noted if known.

2.3.0 Short-rotation and seasonal cropping #C6A14D

This class includes agricultural practices characterised by the cultivation of crops with rapid turnover cycles and seasonal planting patterns, typically optimised for efficient land use and high-yield production.

Scope note This class encompasses agricultural activities such as the cultivation of fast-growing crops such as maize, barley, and certain vegetables, often rotated on short-term cycles to maintain soil health and maximise productivity.

Scope note Arable, and mixed livestock cropping, integrating both crop cultivation and animal husbandry, are also included in this class.

Scope note The period of rotation to be considered (e.g. in the listing of commodities, or when deciding whether there is any rotation at all) is five years.

Usage note If the type of crop is known, record this in the commodity field. For mixed arable and livestock operations, both commodities should be captured as the primary commodity (neither is ancillary).

Change note The explicit five year periodicity was added in v0.4.0.

Change note https://github.com/manaakiwhenua/nzsluc/issues/65

2.3.1 Arable cropping #A98734

Predominantly grain, seed, or fodder crops (and over time may include vegetable rotations). May be used for dairy support.

Usage note The specific commodity or commodities should be noted if known.

2.3.2 Arable and mixed livestock cropping #C6A14D

Predominantly grain, seed, or fodder crops; over time it may include short-term (c. 1–3 years) pasture and livestock rotations, and/or vegetable rotations. Pasture and livestock rotations may occur less than 50% of the time..

Scope note Includes maize, barley, wheat, peas, other grain and seed crops, and fodder crops. May be used for dairy support (which should be recorded as a management practice, if this is known).

Usage note The specific commodity or commodities should be noted if known.

2.3.3 Short-rotation horticulture #E3BC67

Crop plants living for less than 2 years that are intensively cultivated; predominantly rotations of vegetable crops or seasonal fruits grown for human consumption.

Scope note May include livestock rotations, but this is considered less likely

Usage note The specific commodity or commodities should be noted if known.

2.3.4 Seasonal flowers and bulbs, and turf farming #FED57E

Cultivation of seasonal ornamental flowers, bulbs, and turf grass for commercial purposes.

Scope note This class encompasses activities such as the cultivation of flowers and bulbs for seasonal markets, landscaping, and turf-farming for sports fields, lawns, and recreational areas.

2.4.0 Perennial horticulture #9EA083

Crop plants living for more than 2 years that are intensively cultivated, usually involving a relatively high degree of nutrient, weed, and moisture control.

Usage note Management practices such as irrigation or netting can be recorded as management practices.

Usage note Crop type can be recorded as the commodity and used to identify key crops (e.g. kiwifruit, avocados, and grapes).

2.4.1 Tree crops #818367

Long-term cultivated plants, typically trees or woody shrubs, grown for fruits, nuts, or other edible parts. These crops require intensive management practices aimed at ensuring optimal growth, productivity, and quality over multiple years.

Usage note Examples include orchard fruits such as olives, apples, oranges, and apricots, as well as tree nuts such as hazelnuts, macadamias, and chestnuts, which should be recorded in the commodity attribute, if known.

Usage note Management practices may include pruning, pest and disease control, irrigation, and harvesting techniques specific to tree crops.

2.4.2 Vine fruits #9EA083

Fruit-bearing plants that grow on vines or trailing stems, often requiring support structures (e.g. trellises, arbours). These plants produce fruits that typically hang from vines and may include grapes, kiwifruit, and passionfruit. Vine fruit cultivation involves specific management practices such as pruning, training, trellising, and pest and disease control specific to vine plants.

2.4.3 Other perennial horticulture #BBBEA0

Perennial plants beyond tree crops and vine fruits, such as berries (e.g. strawberries, blueberries), perennial herbs (e.g. lavender, rosemary), and ornamental perennials (e.g. roses, lilies).

2.5.0 Intensive horticulture #4A4C39

Intensive forms of plant production, often with special-purpose improvements used for horticultural production.

Usage note Record management practices such as hydroponic systems, irrigation, and netting, if known.

Usage note The specific commodity or commodities should be noted if known.

2.5.1 Production nurseries #565126

Specialised intensive horticultural facilities dedicated to propagating and growing plants for commercial purposes (outdoor or under cover).

Scope note These nurseries focus on cultivating a wide range of plant species, including ornamentals, fruit trees, and shrubs, typically for landscaping, reforestation, or ornamental purposes.

Scope note Nurseries may be outdoor (exposed) or under cover. Production nurseries may employ advanced techniques such as grafting, tissue culture, and controlled environment systems to optimise plant growth and quality.

2.5.2 Glasshouses/shadehouses #797346

Controlled-environment structures for protected cultivation of plants, including vegetable and floriculture production.

Scope note Glasshouses provide a transparent enclosure, typically made of glass or plastic, allowing natural sunlight to penetrate while shielding plants from adverse weather conditions. Shadehouses utilise shade cloth or netting to regulate light exposure and temperature levels. These structures enable year-round production of a wide variety of crops, including vegetables, herbs, flowers, and ornamental plants.

2.6.0 Intensive animal production #C79588

Intensive forms of non-pastoral animal production, livestock production facilities, or animal-holding yards.

Usage note Record animal types as commodities, if known.

Usage note The management field can be used to record practices such as free range or organic.

2.6.1 Intensive animal containment #9A6C60

Facilities or systems designed to confine and manage animals, including feed lots, pens, dairy sheds and yards, and herd homes, where animals are raised intensively for meat production or other purposes.

2.6.2 Poultry farms #C79588

Specialised dacilities dedicated to the intensive production of domestic fowl (e.g. chickens, turkeys, ducks, geese) for meat or egg production.

2.6.3 Piggeries #B08074

Facilities designed for the intensive rearing of pigs.

2.6.4 Horse studs #DDAA9D

Establishments focused on the breeding and raising of horses.

Usage note Applies to the intensive part of horse studs; relatively more extensive and pastoral aspects are not included.

2.6.5 Aquaculture #F5C0B2

Controlled cultivation of aquatic organisms (e.g. fish, molluscs, algae) in natural or artificial environments, typically for food production, but also for stock feeds, pharmaceutical uses, biofuels, etc.

2.7.0 Water and wastewater #528582

Built water features associated with agricultural use.

2.7.1 Stock water #528582

Reservoirs or farm dams on agricultural land for the purpose of supplying drinking-water for stock.

2.7.2 Effluent pond #205351

Effluent ponds typically associated with dairying.

2.7.3 Water treatment – land application #396B69

Land used for effluent disposal

> **Scope note** probably an ancillary use where some form of grazing is the primary use.

2.7.4 Water treatment – wetland #6C9F9C

Constructed or natural wetlands used to improve water quality prior to discharge.

2.7.5 Irrigation reservoirs and canals #86BAB7

Land used for water storage, management, or distribution intended for agricultural purposes; areas allocated for irrigation for agricultural purposes.

2.8.0 Vacant and transitioning land #B6C8BC

As a subclass of Production agriculture and plantations, land must recently have been used for agriculture or plantations.

Scope note In the case of greenfield development where land use is transitioning to a built-environment category, classify under 3.9.0 Vacant and transitioning land (or a subclass) rather than 2.8.0.

Change note https://github.com/manaakiwhenua/nzsluc/issues/36

Change note Preferred label changed from 'Land in transition' at v0.4.0

2.8.1 Rehabilitating land #B6C8BC

Land in the process of rehabilitation for agricultural production (e.g. after significant flooding), and which is actively being recovered.

Change note Changed notation at v0.4.0

2.8.2 Abandoned land #CADCD0

Land where a previous pattern of agriculture may be observed but that is not currently under production, but not due to physical land degradation.

Change note Changed notation at v0.4.0

2.8.3 Degraded land #94A599

Unused land that is degraded through erosion or flood events that is not being rehabilitated.

Usage note Can include contaminated land.

Change note Changed notation at v0.4.0

2.8.4 No defined use #DBEDE0

Land cleared of vegetation where the current proposed land use is unknown.

Example Land being prepared for new primary production use (farms, mines, etc.) but where the precise nature of the proposed land use is unknown.

Change note Changed notation at v0.4.0

2.9.0 Smallholder farm #E5D64F

A rural property larger than a rural residential property (> 2 ha) but where the primary motivation is still residential rather than commercial agricultural production, and where any land management is low-intensity and incidental to the residential occupation.

Scope note Extends 3.1.4 Rural residential to properties that exceed the lifestyle block size threshold but retain distinctive residential/lifestyle character. Distinguishable from 2.2.0 Grazing modified pasture systems by the absence of commercial pastoral intent and the relatively low level of active land management.

Scope note Properties classified here may carry small numbers of stock or maintain informal pasture, but these activities are subsidiary to the residential use of the land.

Usage note As a subclass of 2.0.0, commodities and management practices may still be recorded.

Usage note Typical evidence: NDVR lifestyle codes (property uses 21 single-unit lifestyle, 22 multi-unit lifestyle) or possibly category L with land area in the range 2–40 ha. Properties substantially exceeding 40 ha with lifestyle NDVR codes should typically be excluded, in favour of classification under 2.2.2 or 2.2.3.

Change note Added at v0.4.0 to capture the gap between 3.1.4 Rural residential (0.4–2 ha) and commercial pastoral classes (2.2.0).

3.0.0 Built environment #8E7A78

Land uses that involve high levels of interference with natural processes, generally in association with the built environment of closer settlement and supporting infrastructure. The level of intervention may be high enough to completely remodel the natural landscape – the vegetation, surface and groundwater systems, and land surface.

Scope note Active outdoor recreation and/or tourism outside of urban areas that is not captured in other land-use categories may be captured under 3.2.1 Outdoor recreation, though may still retain much natural character.

3.1.0 Residential #815E5B

Land uses primarily designated for private, long-term human habitation, encompassing a variety of housing types and densities.

Scope note The class includes holidays homes, retirement villages, student hostels and other forms of low-turnover or longer-term accommodation, even if these are run commercially. Shorter- term forms of accommodation (motels, hotels, holiday parks, etc.) are captured under ‘3.3.0 Commercial’ or ‘3.3.3 Hospitality’

Usage note Some alignment with the National Planning Standards (Zone Framework Standard) (MfE 2019) has been considered in the development of the subclasses. This information (as at time of classification) should be captured in the zone attribute to allow for some reclassification and specification.

3.1.1 High-density residential #4C3B39

Areas characterised by a high concentration of housing units per unit of land area, typically in multi-storey buildings or high-rise developments, often found in urban centres supporting high population density.

3.1.2 Medium-density residential #644E4C

Areas featuring a moderate concentration of housing units per unit of land area, often in suburban or semi-urban settings.

Scope note Alignment to the Medium Density Residential Standards (MfE 2022) should be considered.

Example Townhouses, terraced housing, low-rise apartments.

3.1.3 Low-density residential #815E5B

Residential properties within urban boundaries that fall within large-lot or low-density residential zones and are often single-family, one- to two- storey houses with yards and landscaping and lower population density.

3.1.4 Rural residential #A87A76

Residential properties with low-intensity (non-commercial) land management practices on land in rural or peri-urban areas.

Scope note Typically featuring larger parcel sizes amidst agricultural or natural surroundings. Concordant with the ‘Rural lifestyle zone’ from the Zone Framework Standard.

Usage note As a general guideline, properties with a dwelling, and between 0.4 ha and 2 ha in size, should be considered rural residential. Larger properties should be considered under class 2, especially if the zone context is non-residential, and even if they are declared residential or lifestyle in the District Valuation Roll.

Usage note Properties in this size interval and without a dwelling could be in transition to residential.

Historical note In land fragmentation analyses, parcels greater than 2 ha in size (with or without a dwelling) have been considered non-residential. The term lifestyle is disfavoured in this context, as it is not a landuse but rather it indicates that some form of primary production does occur but that it is not the resident's primary source of income. Thus the term 'lifestyle' reflects a motivation for a particular land use, but is not itself a useful land use.

3.2.0 Public recreation and services #C19878

Land designated for recreational facilities and community amenities, serving recreational needs and essential functions of the local population.

3.2.1 Outdoor recreation #C19878

Land areas dedicated to leisure activities conducted in natural or semi-natural settings, typically with considerable unsealed vegetated areas.

Scope note Parks or reserves with a high level of native bush or that are protected areas should be classified under class 1.

Scope note The specific identification of this land is intended to enable more ready identification of urban green space. However, this category may also be used to identify developed recreational areas that fall outside urban boundaries, such as mountain-bike parks.

Example Parks, trails, sportsgrounds, camping grounds, zoos, botanic gardens, recreational reserves.

3.2.2 Indoor recreation #958273

Facilities designed for recreational activities conducted within enclosed or semi-enclosed structures.

Example Sports centres, gyms, swimming pools, indoor arenas.

3.2.3 Community services #E09962

Land used for providing essential services and facilities to support the local community.

Example Educational institutions, public healthcare facilities, libraries, museums, courts, prisons, civic buildings, emergency services, marae, religious buildings, cemeteries.

3.3.0 Commercial #6E94BA

Land in a built environment context used for private-sector economic activities, including retail, office, hospitality, entertainment, healthcare, transportation, and warehousing.

Usage note Central government offices when these are privately rented should be captured in this category, even if they are public-facing.

3.3.1 Retail #6E94BA

Areas used for the sale of goods and services directly to consumers.

Example Shops, supermarkets, shopping malls, convenience stores, suburban dairies, and other retail outlets.

3.3.2 Office #396285

Land and buildings primarily used for administrative, professional, or managerial activities.

Example Corporate offices, professional services, banks, government buildings, business parks, coworking spaces.

3.3.3 Hospitality #A4CAF2

Land and buildings providing accommodation, food, and beverage services to the public.

Example Hotels, motels, resorts, bed and breakfast establishments, restaurants, bars, nightclubs.

3.3.4 Entertainment #89AFD6

Land areas and facilities offering leisure and recreational activities for public enjoyment.

Example Cinemas, theatres, amusement parks, casinos, concert venues.

3.3.5 Healthcare #D0E5FC

Land dedicated to providing private medical services aimed at diagnosis, treatment, and prevention of illness or injury.

Example Hospitals, specialist clinics, physiotherapists, medical offices, laboratories

3.3.6 Transportation & warehousing #547B9F

Privately owned land associated with the movement and storage of goods, including distribution centres, warehouses, logistics facilities, car dealerships, and commercial parking lots.

Scope note Excludes airports, ports, roads, and train stations (captured under 3.6.0)

3.4.0 Manufacturing and industrial #9598A8

Land uses dedicated to production, processing, and industrial activities, contributing to the production of goods and materials for commercial purposes.

3.4.1 General purpose factory #9598A8

An area used for manufacturing, assembly, or repairs of various products.

Scope note Includes some specialised or purpose-built machinery and equipment for mass production across multiple industries.

3.4.2 Food processing factory #CED1E2

An area where the principal use is food processing, packaging, and preservation. The building is most likely to be purpose built and may have extensive plant and equipment included.

Example Cannery; milk production plant.

3.4.3 Major industrial complex #838695

An area with large-scale industrial use and significant infrastructure .

Example Car plant, paper mill.

3.4.4 Sawmill #A8AABB

An area with special improvements for the processing (milling and curing) of raw timber, wood products, and by-products.

3.4.5 Farm buildings/infrastructure #E2E4F6

An area with buildings, sheds, and other infrastructure associated with farm enterprises supporting agricultural operations.

Scope note If it is possible to discriminate a domestic area from a farm, the house and adjoining non-productive land should be classified as 3.1.4 Rural residential.

Example Barns, silos, storage sheds, and irrigation systems.

3.4.6 Abattoirs #BBBDCE

Areas with specialised improvements for the slaughter of stock and the preparation of meat for the wholesale market.

3.5.0 Utilities #955085

Land allocated to providing electricity, gas, or water; or for navigation and communication infrastructure.

Change note https://github.com/manaakiwhenua/nzsluc/issues/69

Change note Class 3.5.0 redefined at v0.4.0 to include naviation and communciation infrastructure, which was formerly captured under class 3.6.0.

3.5.1 Fuel powered electricity generation #7652A3

Facilities that produce electricity by burning fossil fuels such as coal, oil, and natural gas.

3.5.2 Hydroelectricity generation #9670C4

Facilities that use the energy of flowing or falling water (typically hydroelectric dams) to generate electricity.

Scope note Includes canals and other water conveyance infrastructure associated with hydroelectricity generation.

3.5.3 Wind electricity generation #D6B2FD

Power generation from wind, including wind farms.

3.5.4 Solar electricity generation #B78FE5

Facilities that harness sunlight using photovoltaic cells or solar thermal systems to generate electricity.

3.5.5 Electricity substations and transmission #955085

Facilities and infrastructure associated with distribution and transmission of electrical power.

Example Substations, transformers, high-voltage transmission towers.

3.5.6 Gas treatment, storage, and transmission #B36CA2

Facilities and infrastructure for processing, storage, and transport of natural gas.

Example Treatment plants, storage facilities, pipelines.

3.5.7 Water extraction and transmission #D289BF

Facilities and infrastructure for extracting, purifying, treating, and transporting water from natural sources to meet human needs.

Scope note Includes drinking water reservoirs.

3.5.8 Navigation and communication #F2A6DE

Navigation and communication infrastructure.

Example Radar stations, beacons, lighthouses, TV and radio transmission towers.

Change note https://github.com/manaakiwhenua/nzsluc/issues/69

Change note Notation changed from 3.6.5 to 3.5.8 at v0.4.0 when class 3.6.0 redefined to exclude navigation and communication infrastructure.

3.6.0 Transportation #72737E

Land allocated to infrastructure used for the transportation of goods or people.

Usage note Zoning of land for roading or rail purposes that has not yet been constructed should be captured as attributes.

Change note https://github.com/manaakiwhenua/nzsluc/issues/69

Change note Redefinition of class 3.6.0 to exclude navigation and communication infrastructure, which is now captured under 3.5.8. Renamed from 'Transport and communication'.

3.6.1 Airports/aerodromes #75777E

Locations from which aircraft flight operations take place, including areas used for the accommodation of aircraft and coordination of air cargo or passengers (including heliports).

3.6.2 Roads #494B51

The full roading corridor, including sealed road areas and unsealed roadside strips.

3.6.3 Railways #33353A

Permanent rail transport tracks and associated infrastructure, including stations and terminals; includes the full rail corridor required for safety clearance.

3.6.4 Ports and water transport #61636A

Harbour locations where ships dock to transfer people or cargo, including terrestrial parts and designated marine areas that are specifically designated and controlled for activities associated with the port.

Usage note The water attribute can be used to differentiate the terrestrial and marine parts of the same facility.

Example Ports, docks, and wharves.

Example Harbours, navigational channels.

3.7.0 Mining #E1CEA7

Mining and extractive industries (including salt extraction from evaporation basins).

Usage note Record the specific type of mining activitiy in the commodity attribute, if known.

3.7.1 Mines #836639

Land from which minerals, precious stones or coal is being extracted, including open-cut and deep-shaft mines.

3.7.2 Quarries #9E7E51

Land from which stone, gravel, clay, slate, sand, soil, rock, or other construction materials are being extracted.

3.7.3 Tailings #BA9869

Tailings dumps and dams for storage or treatment of waste material left over after extraction, often stored in containment facilities or ponds (including quarrying output).

3.7.4 Evaporation basins #FCD7A5

Basins and associated facilities used for evaporation of water from irrigation drainage or (predominantly) salt extraction.

3.7.5 Extractive industry not in use #D7B383

Land undergoing rehabilitation after mining activities, and unmined land within a mining permit.

3.8.0 Waste treatment and disposal #E77148

Land uses dedicated to managing and processing waste materials (solid waste, wastewater, stormwater), primarily associated with industrial and urban activities, to mitigate environmental impacts and protect public health.

3.8.1 Landfills #AC3F25

Designated areas for disposal of solid waste, where waste materials are deposited, compacted, and covered to reduce contamination and control emissions.

Scope note Landfill gas recovery systems are classified as part of a landfill land use.

Scope note Includes all class 1 to 5 landfills

Example Landfills designed to receive inert construction materials.

3.8.2 Transfer stations and recycling facilities #E97453

Facilities where solid waste is collected, sorted, processed, and prepared for recycling or transfer to landfills or other disposal sites.

3.8.3 Municipal wastewater #CB593B

Wastewater generated from residential, commercial, and industrial activities within urban areas requiring treatment before discharge or reuse; includes municipal wastewater ponds and sewerage pipelines.

3.8.4 Wastewater treatment – land application #FF9475

Areas where treated municipal wastewater is applied onto land for beneficial reuse, following appropriate treatment processes to ensure environmental safety.

Usage note This is often likely to be an ancillary use.

Example Irrigation, groundwater recharge, soil fertility enhancement.

3.8.5 Stormwater management #FEBDAA

Infrastructure aimed at controlling and mitigating stormwater runoff to prevent flooding, erosion, and pollution.

Example Detention basins, drainage systems, retention ponds, rain gardens, wetlands.

3.9.0 Vacant and transitioning land #D69B97

This land-use type includes areas that are currently unused or undergoing a transition from one land use to another, but in this case with a clear transition towards or within other concepts under the built environment.

3.9.1 Brownfield development #87636E

Areas of active redevelopment of previously developed (often industrial) land that may be abandoned, contaminated, or under-utilised, to rehabilitate and repurpose sites for new urban activities.

Scope note May include residential areas undergoing infill development that increase housing density.

Change note Notation changed from 3.9.3 to 3.9.1 at v0.4.0

3.9.2 Greenfield development #AC8792

Previously undeveloped or agricultural land zoned for or undergoing new construction projects or urban expansion, involving conversion of rural or natural areas into built uses.

3.9.3 Idle and derelict land #D3ACB7

Derelict land and developed land that is idle.

Change note Notation changed from 3.9.1 to 3.9.3 at v0.4.0